THE FLOOD AND CHALLENGES OF RASUWAGADHI, KERUNG BORDER

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Abstract

On July 8, 2025, a glacial lake outburst flood (GLOF) originating from the Puruogangri Glacier in Tibet caused severe damage in Rasuwagadhi, a major customs checkpoint on the Nepal - China border. The event destroyed critical infrastructure, including the Nepal - China Friendship Bridge, highways, and four hydropower plants, resulting in the loss of approximately 8 - 9% of Nepal's electricity generation capacity. At least nine fatalities and nineteen missing persons were reported. This study draws on meteorological records, satellite imagery, and secondary disaster reports to determine the causes and consequences of the flood. Analysis confirmed the absence of unusual precipitation, instead linking the disaster to abnormal glacial melting and rapid expansion of a supraglacial lake in 2024 - 25. Comparative cases from Nepal, India, and Peru demonstrate the recurrent and global nature of GLOFs in high mountain regions. The findings highlight the role of accelerated climate change in increasing glacial hazards and emphasize the urgent need for systematic monitoring, structural risk reduction, and transboundary collaboration. The Rasuwagadhi case contributes to broader research on Himalayan climate vulnerabilities, underlining that without proactive adaptation, GLOFs will increasingly threaten lives, infrastructure, and regional economies.

Keywords: *Glacial lake outburst flood (GLOF); Himalayan cryosphere; climate change impacts; hydropower vulnerability; transboundary water governance; disaster risk reduction; remote sensing; high mountain hazards*

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BACKGROUND

Rasuwagadhi is known as a major Himalayan customs checkpoint located in Rasuwa district of Nepal. It lies along the newly constructed Rasuwagadhi - Kerung road, which directly connects Kathmandu with Tibet. This road serves as an important trade route between Nepal and China. Local communities live in the hilly terrain at an elevation of about 2,800 to 3,000 meters, sustained by the Bhotekoshi River that gives life to the region.

As Nepal’s second official border checkpoint with China, this location carries both economic and geopolitical importance (Figure I shows this border before the devasting flood). The Lhende and Kerung rivers of Tibet are the main sources shaping the Bhotekoshi Valley. Since these Himalayan rivers flow with great force, the risk is ever-present.

Figure I Kerung Border before flood, Source: RSS, 2023

Objectives

This report main objective are pointed below:

1. To discuss the causes and effects of the July 8, 2025 Rasuwagadhi flood.
2. To discuss the impact of climate change and review the event in relation to other GLOFs in Nepal and elsewhere.
3. To advise essential steps to reduce disaster risk through tracking, preparation, and collaboration.

Research Questions

1. How can roads, bridges, and other infrastructure be designed to withstand Glacial Lake Outburst Floods?
2. How can Nepal and China cooperate most effectively to manage and reduce glacier-related hazards?

3. How can glacier, river, and satellite monitoring systems provide early warning to reduce GLOF impacts in the Himalayas?
4. What are the economic and environmental impacts of GLOFs compared to the cost of implementing science-based prevention measures?

Materials and Methods

All the information and data applied in the current paper were collected through secondary sources, and these are peer-reviewed journal publications, reports, and news articles regarding Glacial Lake Outburst Floods (GLOFs) in the Himalayas and concentrating chiefly on the Rasuwagadhi - Bhotekoshi section. Information on flood events, lake and river behavior and observation networks and mitigation actions were abstracted, collated and enriched by information gathered through consultations with colleagues and experts.

The Flood of July 8, 2025

At around 3:15 a.m. on July 8, 2025, a sudden and massive flood struck the Bhotekoshi River. The flood destroyed the Nepal - China Friendship Bridge, swept away customs infrastructure, damaged roads, and caused severe losses to hydropower projects. Hundreds of trucks, electric vehicles, and heavy machinery were reported to have been carried away by the river. Storage warehouses were destroyed, and commercial activities came to a halt (Figure II illustrates the Kerung border after the flood).

This year's monsoon caused major damage along the Galchi - Trishuli - Rasuwagadhi highway. Continuous rainfall triggered landslides, damaging over one kilometer of the highway. Local bridges and slopes along this trade route, which connects Nepal with Tibet, also collapsed. As a result, trade activities between the two countries, as well as local movement, were disrupted. The damage not only halted commerce but also severed contact with the border area.

Figure II Kerung Border after flood, Source: RSS, 2025

In terms of human loss, at least nine Nepalis were killed and nineteen went missing, including some Chinese nationals (Nepal Disaster Risk Reduction Authority, 2025). Initial rescue operations managed to bring more than fifty people to safety.

Four major hydropower projects were affected: Rasuwagadhi (111 MW), Trishuli - 3A (60 MW), Trishuli (21 MW), and Mailung (15 MW). This resulted in a loss of about 8 - 9% of the national power generation capacity (Nepal Electricity Authority, 2025). The Prime Minister was compelled to declare a state of emergency.

Cause of the Flood

The flood was not related to rainfall. Satellite images and meteorological data confirmed that there had been no unusual precipitation (NOAA, 2025; ECMWF, 2025). However, the water level in the upper Bhotekoshi region suddenly rose by 3 - 4 meters, which is typically a sign of a glacial lake outburst. Research indicates that the flood (GLOF - Glacial Lake Outburst Flood) was triggered by the bursting of a supraglacial lake formed in the Puruogangri Glacier in Tibet. Due to abnormal temperatures, the lake had expanded rapidly in 2024 - 25 and eventually burst on July 8 (ICIMOD, 2025).

Discussions

The following discussion explores historical incidents, the danger today, and key mitigation strategies for Glacial Lake Outburst Floods (GLOFs) in the Himalayas and sheds light on key lessons in monitoring, infrastructure, community preparation, and trans-frontier cooperation.

History and Comparison

In Nepal, the Dig Tsho glacial lake outburst in 1985 swept away hydropower projects and hundreds of homes (Ives, Shrestha, & Mool, 2010). In 2021, a flash flood in Chamoli, India, claimed more than 200 lives (Shugar et al., 2021). Similarly, in Peru, a glacial lake outburst in 1941 caused thousands of deaths (Carey, 2010). These examples show that such risks are not limited to the Himalayas but are shared across other mountain regions as well.

Climate Change and Risk

Research has shown that the average temperature in the Himalayas is rising at twice the global rate. This has caused glaciers to melt rapidly, forming new lakes. According to ICIMOD, there are currently about 5,000 glacial lakes in Nepal, which is 14% more than in 1990 (ICIMOD, 2021). This trend is increasing the risk of floods in the coming years.

Lessons and Preparedness: Managing Glacial Lake Risks

The recent sudden flood in the Rasuwagadhi region has highlighted the growing climate risks in the Himalayas and the challenges of managing them. The key lessons drawn from this event are as follows:

1. Monitoring and Early Warning Systems

Glacial lakes are constantly changing. Rising temperatures, melting snow, and unstable underlying rock formations cause rapid fluctuations in water levels and pressure. Continuous monitoring through satellite imagery, aerial photography, and ground-based sensors is therefore essential. If this data is collected in real time and linked to local warning systems, communities at risk can be alerted on time, helping reduce both human and material losses.

2. Lake Management

To prevent sudden glacial lake outburst floods (GLOFs), it is necessary to adopt controlled water release methods. This can be achieved through artificial channels, intake pipe systems, or siphoning techniques. Such management helps balance water pressure within the lake and reduces the risk of sudden outflows.

3. Infrastructure Strengthening

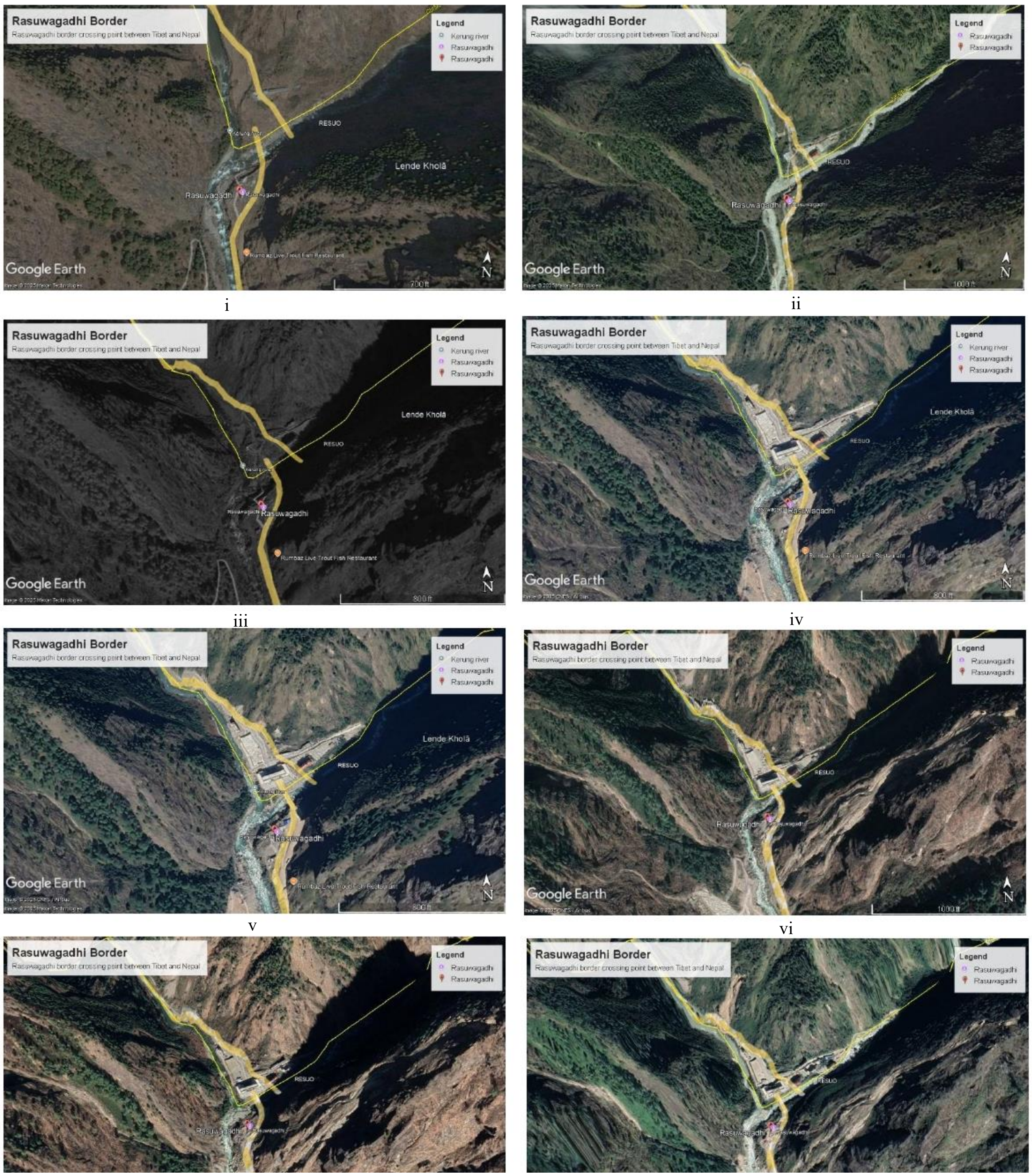

Figure III Satellite images of Kerung border area from 2005 to 2021. i.e. (i) 2005, (ii) 2011, (iii) 2012, (iv) 2014, (v) 2015, (vi) 2017, (vii) 2020, (viii) 2021 Source: Google Earth, 2025

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Roads, bridges, and hydropower projects in mountain regions face high levels of risk. To make these structures sustainable in the long term, additional safety standards must be incorporated into their design. For example, bridges should be built to withstand maximum possible flood levels, road drainage systems should be improved, and hydropower plants should include emergency spillways.

4. **Community Preparedness**

Technical systems alone are not enough; active community participation is crucial in risk management. Local residents must be equipped with emergency response plans, safe site identification, and regular rescue drills. Training schools, women's groups, and local organizations can make disaster response systems faster and more effective.

5. **Cross-Border Cooperation**

Glacial lakes and their risks are not confined to a single country's boundaries. For Himalayan nations like Nepal and China, data sharing, joint monitoring systems, and rapid exchange of emergency information can take risk management to a new level. Such cooperation strengthens scientific research while also creating an environment for immediate assistance during emergencies.

Conclusion

The Rasuwagadhi flood was not just a local incident; it serves as a warning of the increasing GLOF risk across the entire Himalayas. Due to climate change, glacial lakes are expanding rapidly. Without lawful and long-term measures, such disasters will have severe impacts on human lives, infrastructure, and the economy. The message for Nepal is clear: the mountains will bring more floods, and only timely preparedness by society can reduce major losses.

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